

Hand written assignment:-

Q (1) In your words, explain Machine learning differ from traditional "rule-based" programming.

⇒ ML system find pattern automatically by analysing data. You don't need to tell computer how to decide - you give it data and it learn the logic.
Ex. Spam detection.

⇒ In Rule based programming you explicitly define rule/condition (i.e. if amount > 10000.) Basically you tell computer what to do.

Q (2) Q. (a) Supervised learning:- This model uses labeled data which means the correct answer in the training data already known.
Ex. → Email spam detection.

(B) Unsupervised learning:- The model learns from unlabeled data, finding hidden pattern, ~~there~~ without predefined %p.
Ex. → Customer segmentation

(C) Reinforcement learning:- The model where data learns from interacting with env. and gets feedback as rewards/penalties and improve overtime.

Ex. ⇒ Self-driving car



Q.3). Difference b/w labeled and Unlabeled data & which type of ML uses which?

⇒ (i) Labeled Data:- Where each input has a o/p (target / label), we use it in "supervised learning".

Ex. → spam detection

(Some mails can ~~have~~ be tagged as 'spam' as input and some can be 'Not spam'.

(ii) Unlabeled data:- Data which has input but no defined o/p.

Ex. → Customer purchase history ("loyal" & "Churn").
We use it in "Unsupervised learning".

Q.4). Comparison Table: Create a table comparing regression and classification based on:

- Output Type (Continuous & discrete)
- Objective
- Example Use Case.

Aspect	Regression	classification
O/p Type	Continuous value (numeric)	Discrete category (label)
Objective	Predict Quantity	Predict which Category.
Ex.	House price; temperature	Spam detection disease diagnosis



Q. ⑤ Scenario Analysis:- Identify "Regression" or "Classification".

⇒ Predict if an Email is "Spam" or "Not spam".
→ Classification.

⇒ Predict the price of Gold next Monday.
→ Regression

⇒ Predict the exact temperature in Celsius for tomorrow.
→ Regression.

⇒ Categorizing a tumor as "Malignant" or "Benign".
→ Classification.